Geographical distribution of the 25 candidate cities

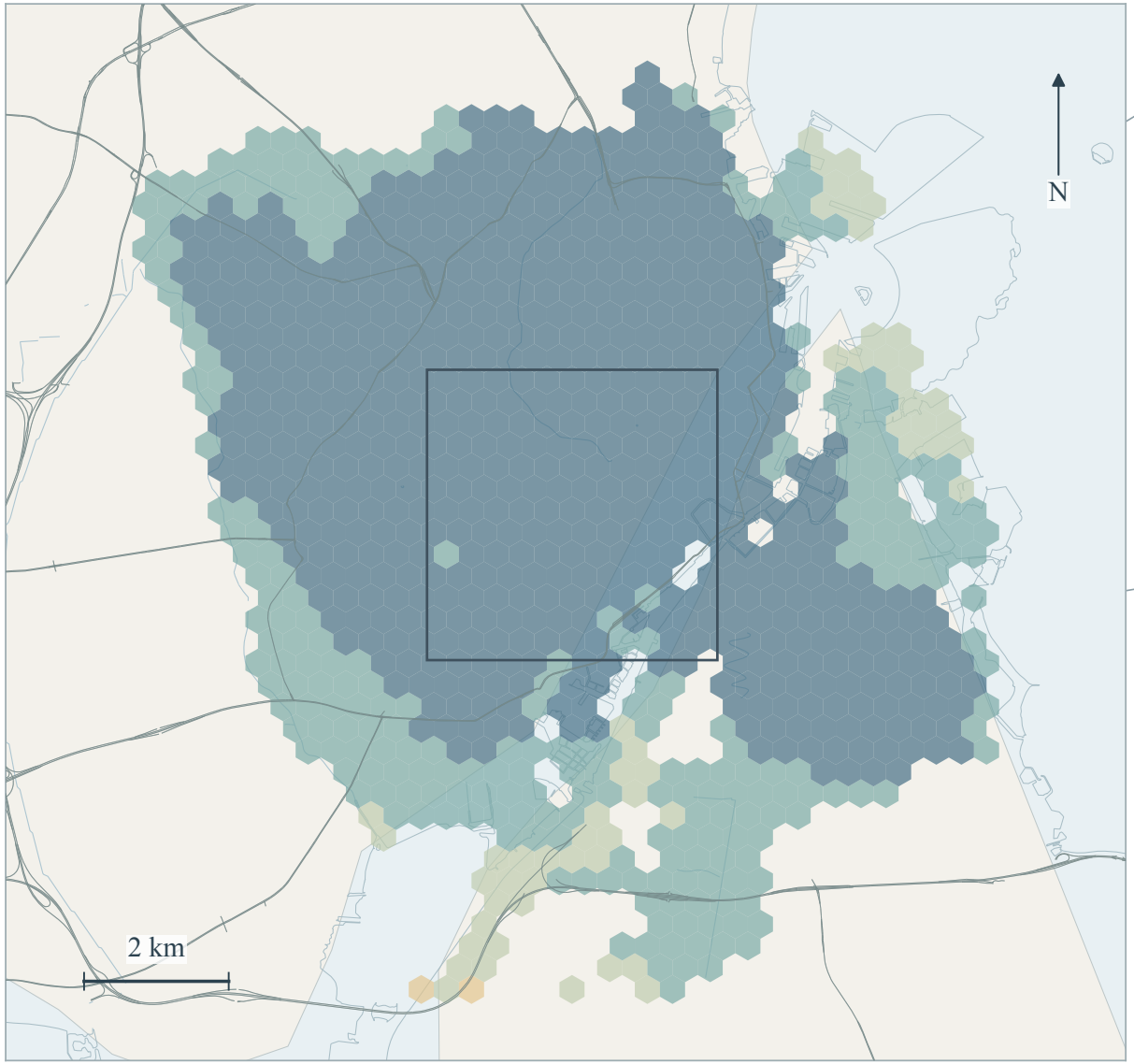
Study locations from the archived city index

25 cities · all candidates shown · Europe enlarged for legibility

Made with Natural Earth | Robinson projection | Urban-area reference locations

Copenhagen

Learning-facility accessibility across retained populated grids



Selected neighbourhood detail

Within 30 min: 99.4% population
Over 60 min: 0.0% population

Modelled walking time (min)

- 0–15
- >15–30
- >30–60
- >60–90
- >90

Selected OSM roads

Mean walking time to 20 nearby learning POIs · Population-weighted summaries
Five time classes · Urban population grids · Independent inset scale

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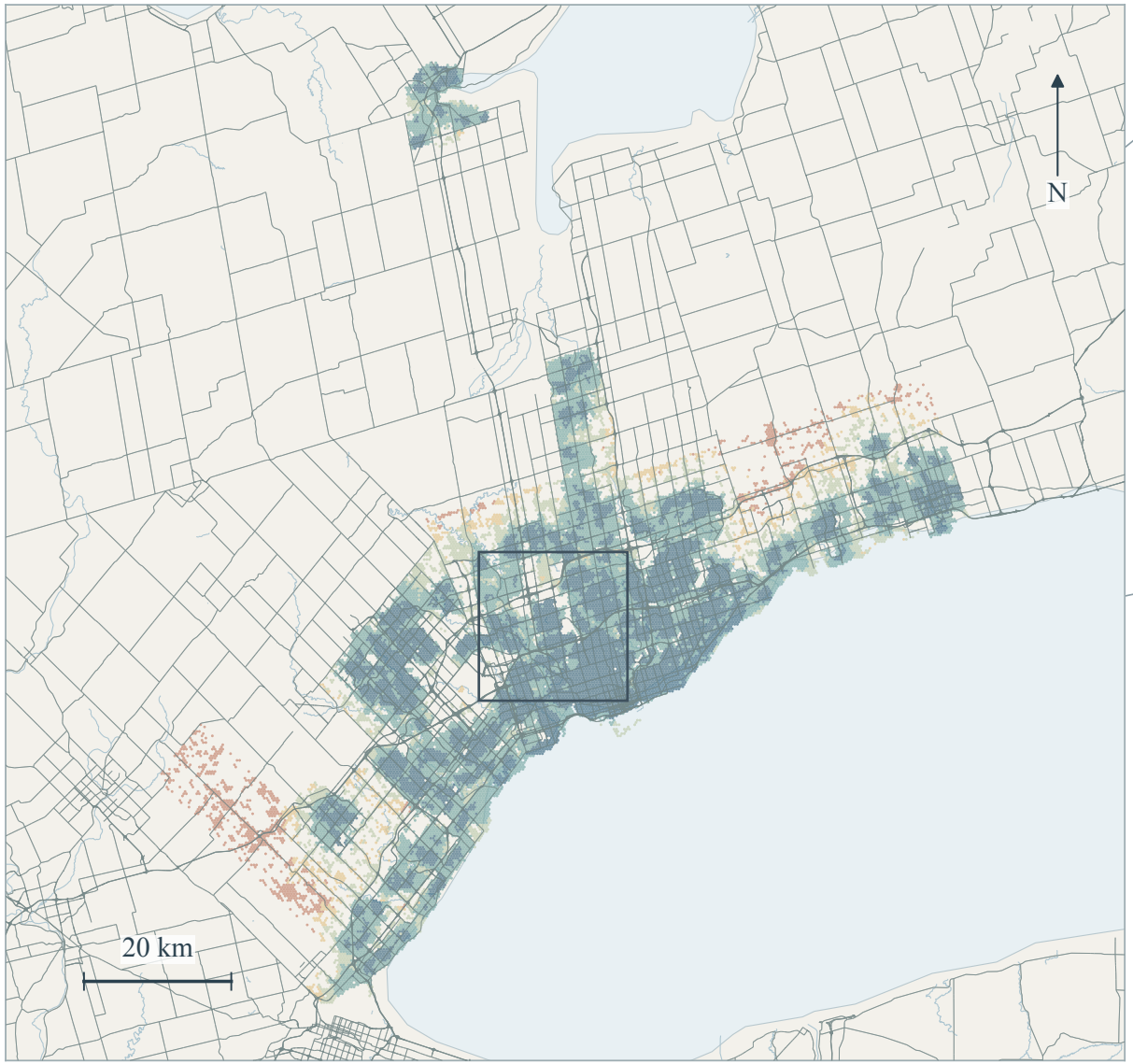
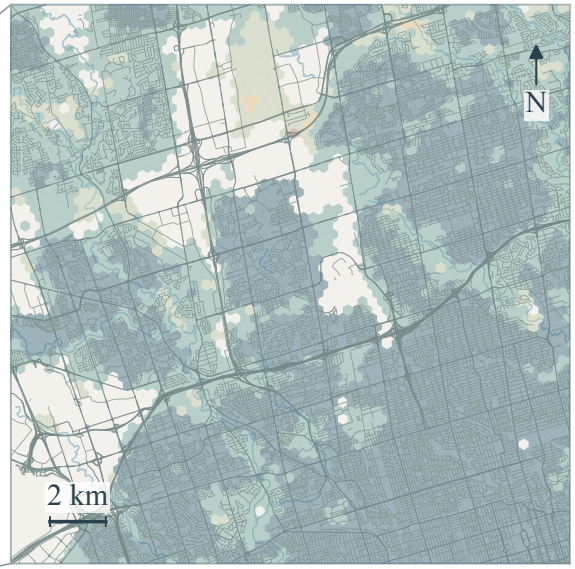
Toronto

Learning-facility accessibility across retained populated grids



Selected neighbourhood detail



Within 30 min: 98.2% population
Over 60 min: 0.3% population

Modelled walking time (min)

- 0–15
- >15–30
- >30–60
- >60–90
- >90

Selected OSM roads

Mean walking time to 20 nearby learning POIs · Population-weighted summaries
Five time classes · Urban population grids · Independent inset scale

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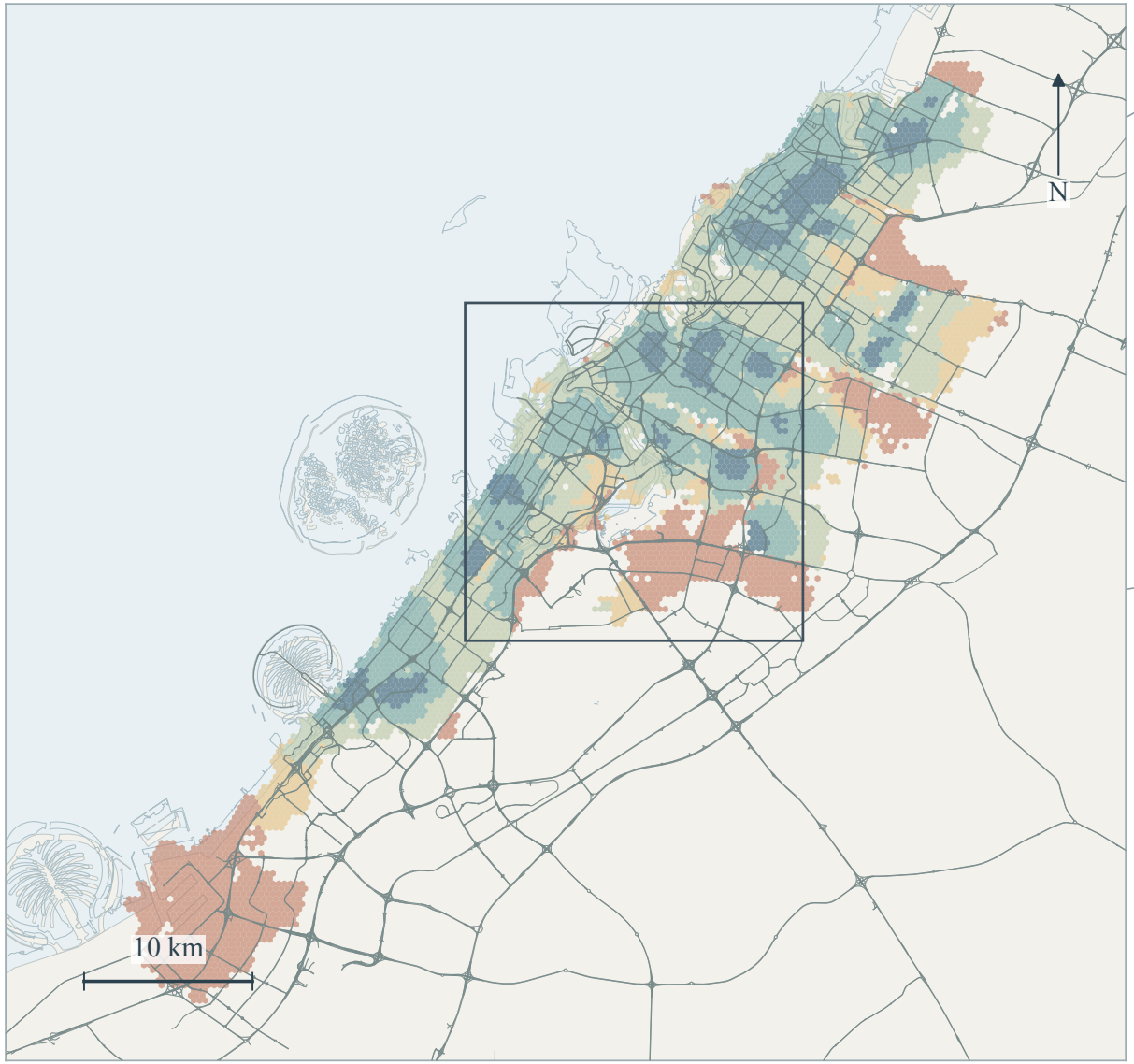
Dubai

Learning-facility accessibility across retained populated grids



Selected neighbourhood detail

Within 30 min: 43.7% population
Over 60 min: 23.4% population

Modelled walking time (min)

- 0–15
- >15–30
- >30–60
- >60–90
- >90

Selected OSM roads

Mean walking time to 20 nearby learning POIs · Population-weighted summaries
Five time classes · Urban population grids · Independent inset scale

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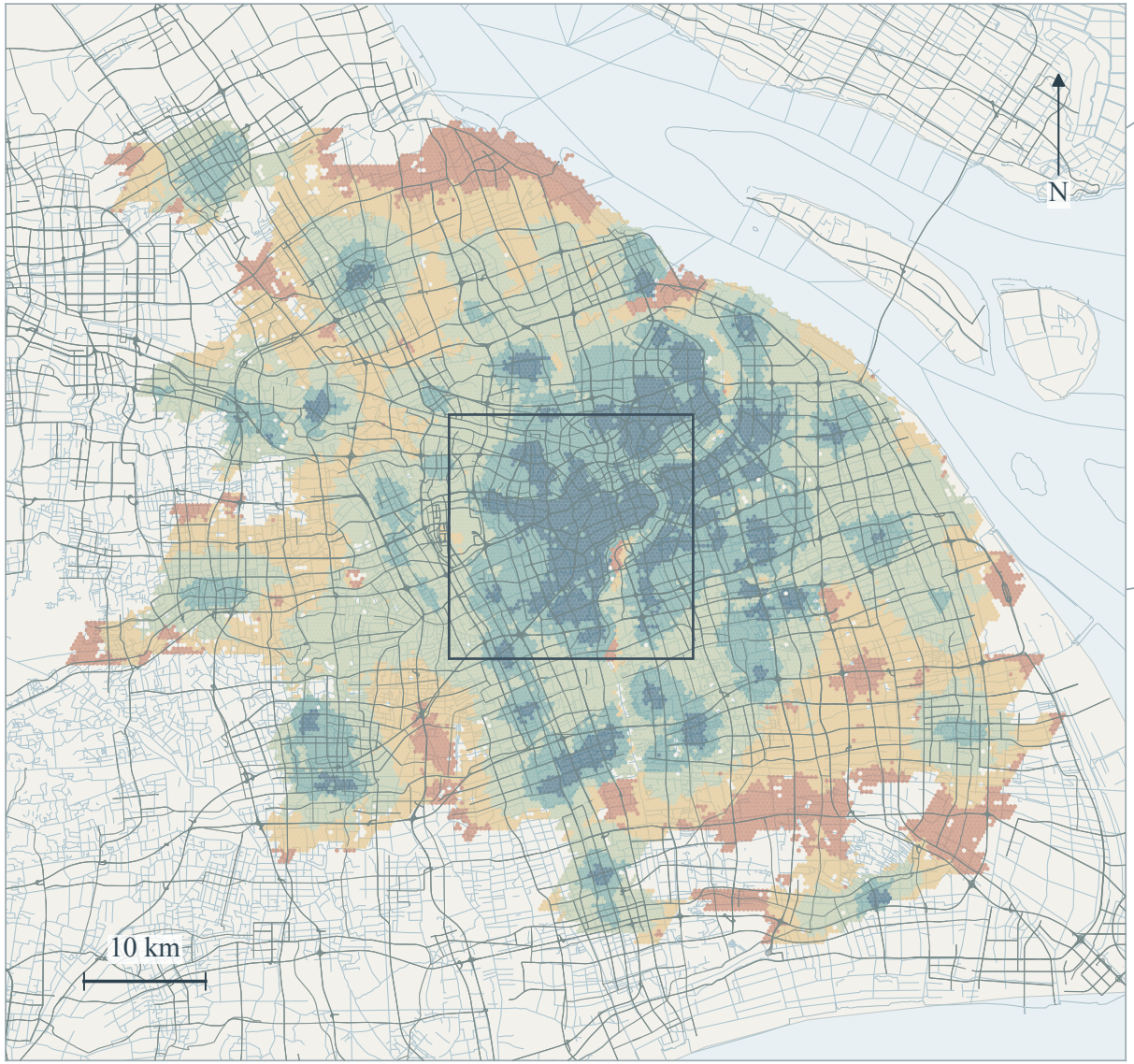
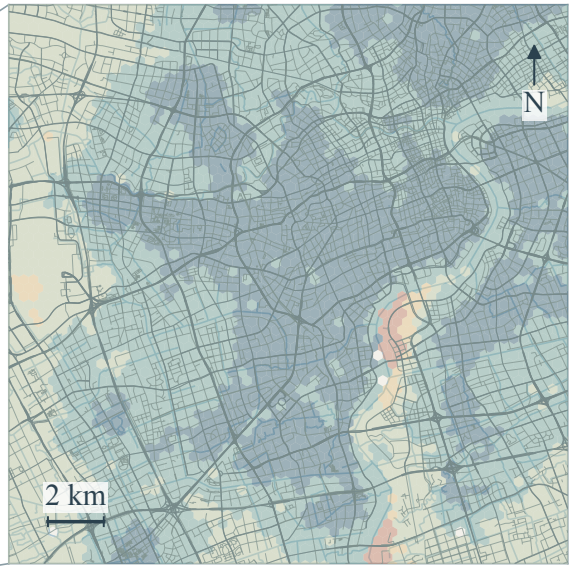
Shanghai

Learning-facility accessibility across retained populated grids



Selected neighbourhood detail



Within 30 min: 58.2% population
Over 60 min: 13.4% population

Modelled walking time (min)

- 0–15
- >15–30
- >30–60
- >60–90
- >90

Selected OSM roads

Mean walking time to 20 nearby learning POIs · Population-weighted summaries
Five time classes · Urban population grids · Independent inset scale

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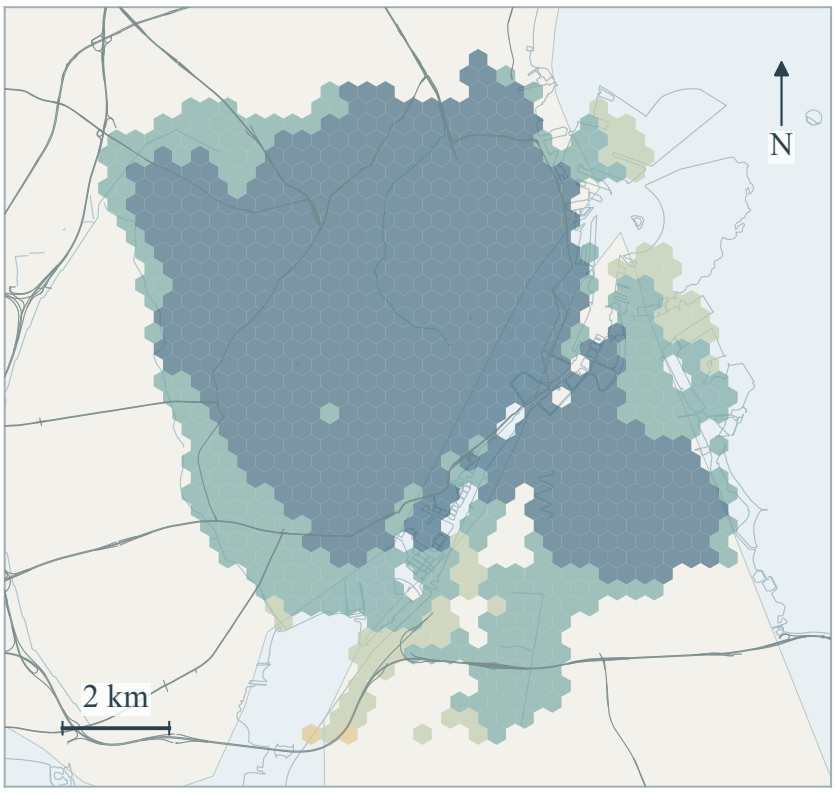
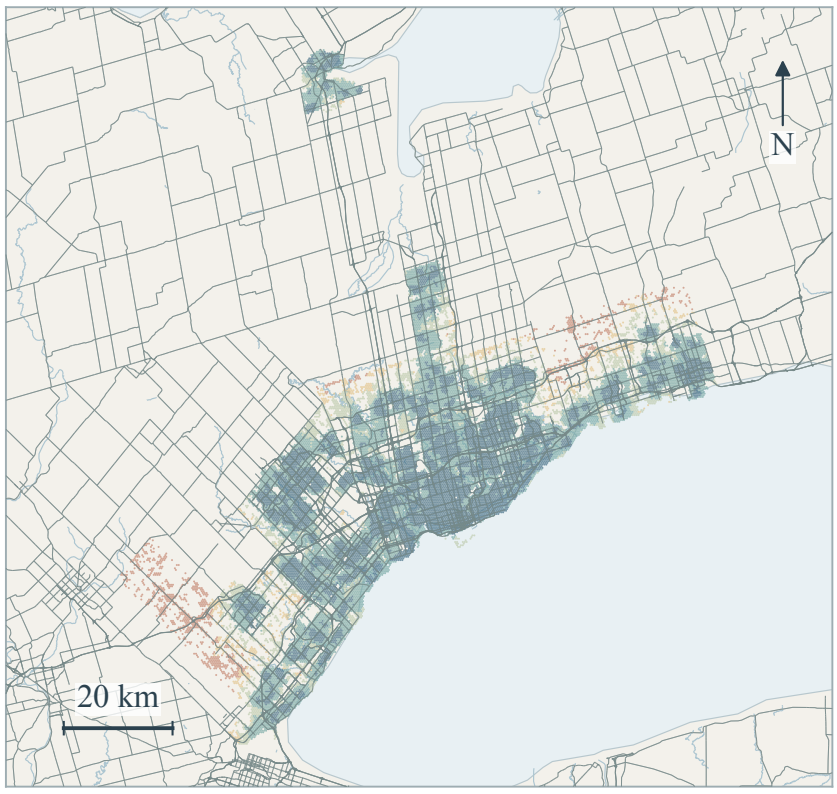
City-scale patterns of learning-facility access

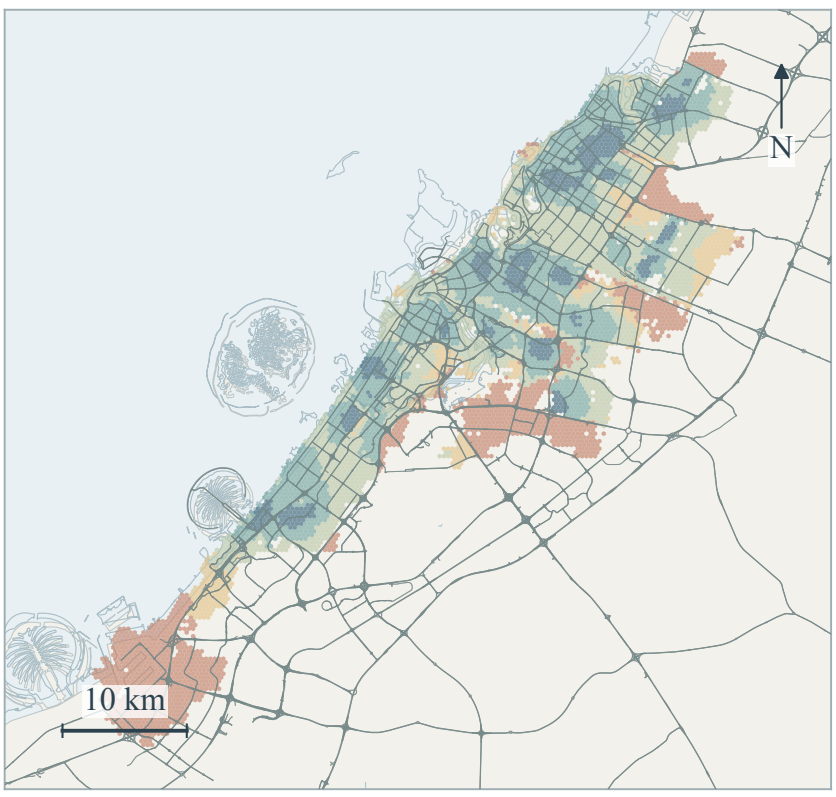
(a) Copenhagen



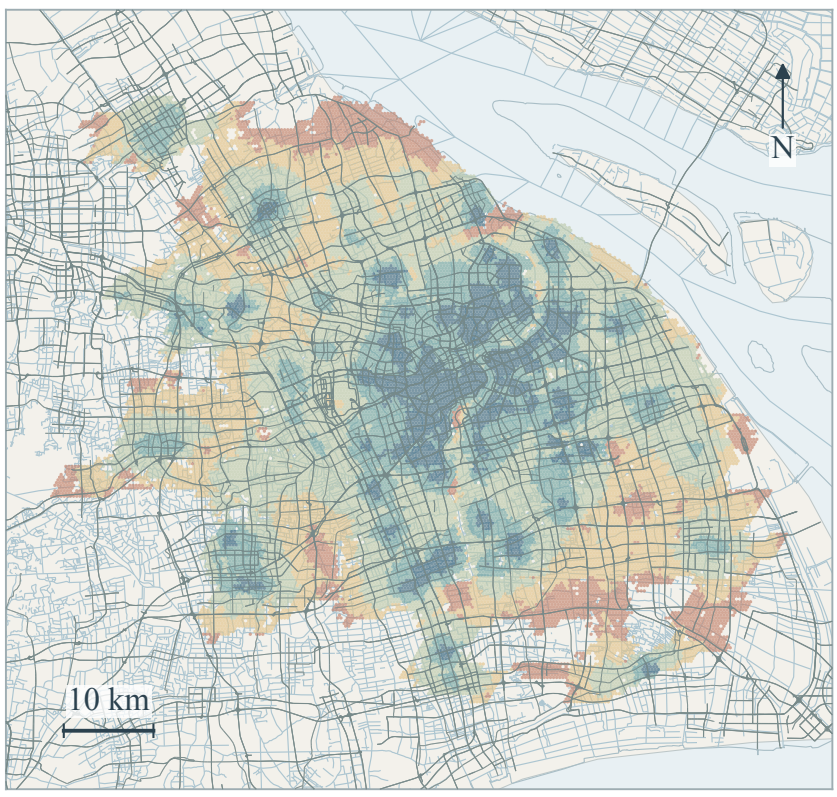
(b) Toronto



(c) Dubai



(d) Shanghai



Modelled walking time (min)

0–15 >15–30 >30–60 >60–90 >90

Shared time classes · City-specific scales · Populated urban grids

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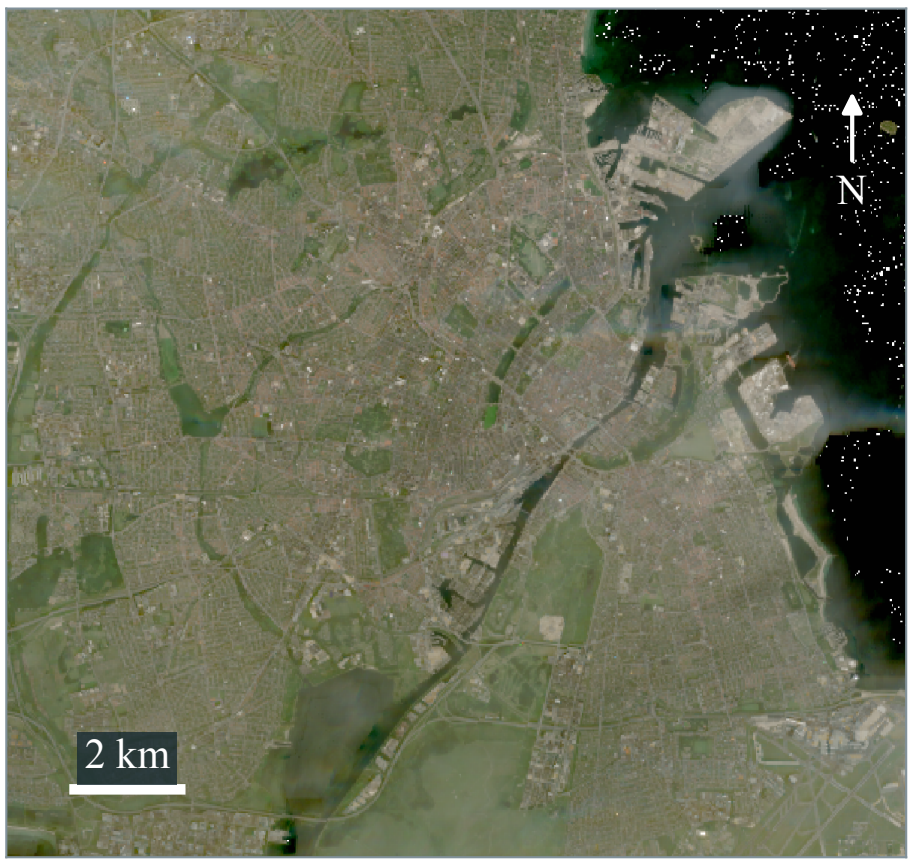
Satellite context for city accessibility

Sentinel-2 / HLS S30 · matched map extents · dated observations

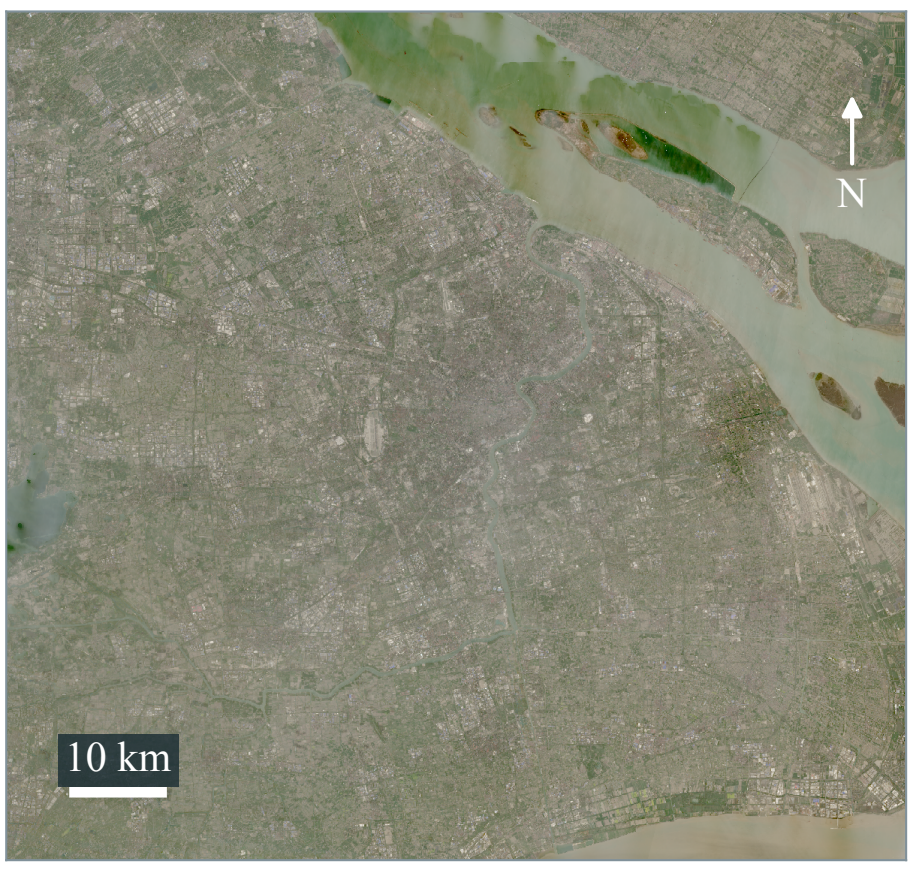
(a) Copenhagen | 2024-06-27



(b) Toronto | 2024-10-17

(c) Dubai | 2024-09-04

(d) Shanghai | 2025-04-07



NASA HLS / GIBS · 30 m source product · EPSG:3857 · city-specific scales

Map captions

Map 1. Geographical distribution of the 25 candidate cities. Locator points follow the archived urban-area index. Coral markers identify the four accessibility case studies; the Europe inset separates nearby labels. Basemap: Natural Earth; world projection: Robinson.

Map 2. Learning-facility accessibility in Copenhagen. Within the retained populated grids, 99.4% of residents have modelled mean walking times to the nearest 20 learning POIs at or below 30 minutes. Colours show five time classes; the inset uses its own scale. Sources: archived Bruno et al. / SONY CSL accessibility data, OpenStreetMap and Natural Earth.

Map 3. Learning-facility accessibility in Toronto. Within the retained populated grids, 98.2% of residents have modelled mean walking times to the nearest 20 learning POIs at or below 30 minutes. Colours show five time classes; the inset uses its own scale. Sources: archived Bruno et al. / SONY CSL accessibility data, OpenStreetMap and Natural Earth.

Map 4. Learning-facility accessibility in Dubai. Within the retained populated grids, 43.7% of residents have modelled mean walking times to the nearest 20 learning POIs at or below 30 minutes. Colours show five time classes; the inset uses its own scale. Sources: archived Bruno et al. / SONY CSL accessibility data, OpenStreetMap and Natural Earth.

Map 5. Learning-facility accessibility in Shanghai. Within the retained populated grids, 58.2% of residents have modelled mean walking times to the nearest 20 learning POIs at or below 30 minutes. Colours show five time classes; the inset uses its own scale. Sources: archived Bruno et al. / SONY CSL accessibility data, OpenStreetMap and Natural Earth.

Map 6. Four-city comparison of learning-facility accessibility. Identical walking-time classes support comparison of spatial patterns; each city has its own scale bar. The population-weighted shares use the retained urban population grids. Sources: Bruno et al. / SONY CSL, OpenStreetMap and Natural Earth.

Map 7. Satellite context of the four accessibility case studies. NASA HLS S30 RGB imagery covers the same geographical extents as the accessibility panels. Acquisition dates and city-specific scales appear on the maps. The source product has 30 m pixels; the displayed images are NASA GIBS browse renderings.